

Cyclone Separator



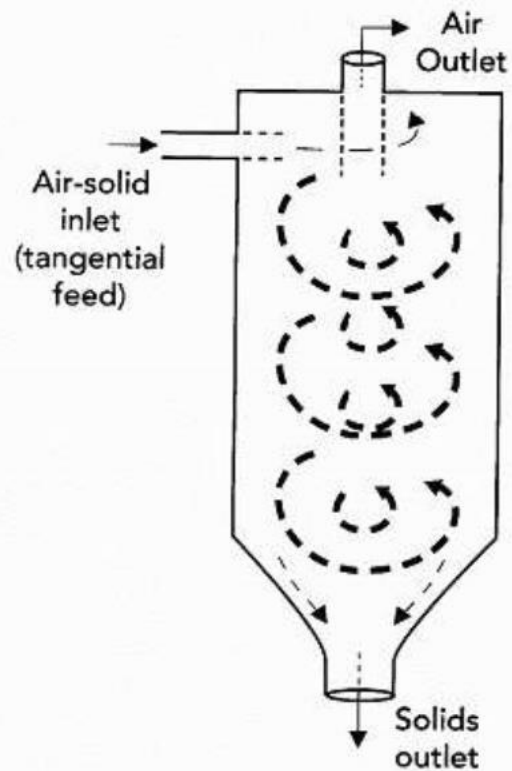
Definition

- Cyclones are mostly used for removing industrial dust from air or process gases. They are the principal type of gas-solid separator
- Most common form of particulate removal gas is rapidly – heavier particulate matter to collect on outside of separator by centrifugal force, where it is collected and removed.

Principles of Operation

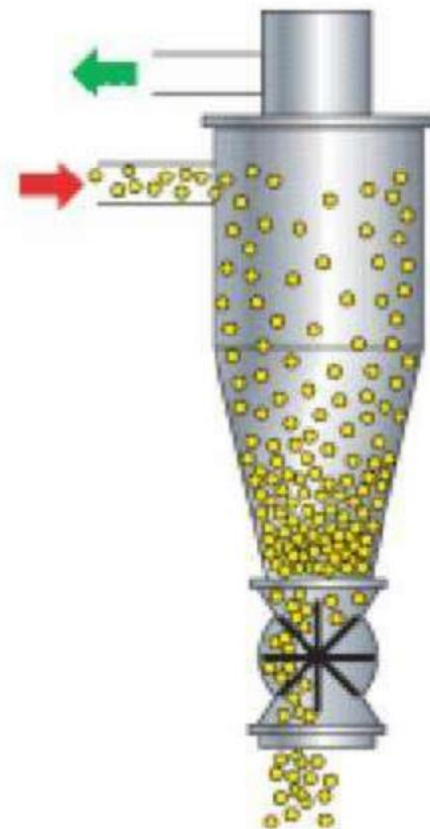
- As with centrifugation, the driving force behind cyclone separation is centrifugal force and the difference in specific gravity between the particle and the carrier gas.
- In a Cyclone, the air or vapor containing particulate material is forced into along the tangential axis. A helical flow pattern is set up within the chamber.
- The centrifugal force causes the particles to migrate to the outside of the chamber. Here they fall down to the bottom of the cyclone by gravity.
- The air moves up the center of the cyclone and reaches the top.

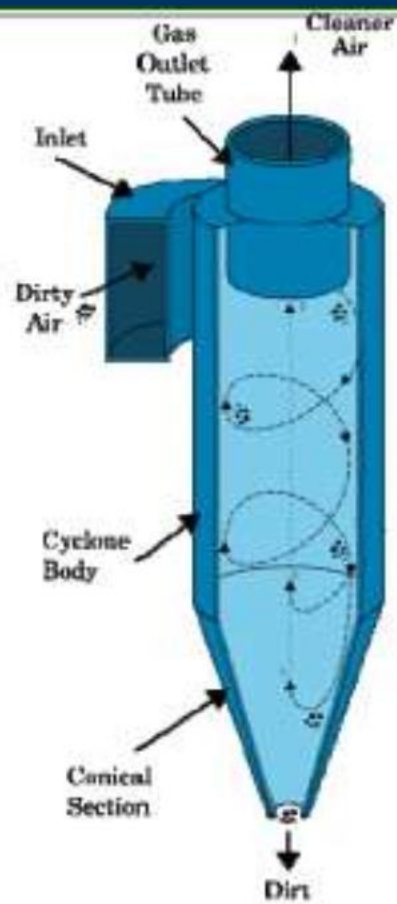
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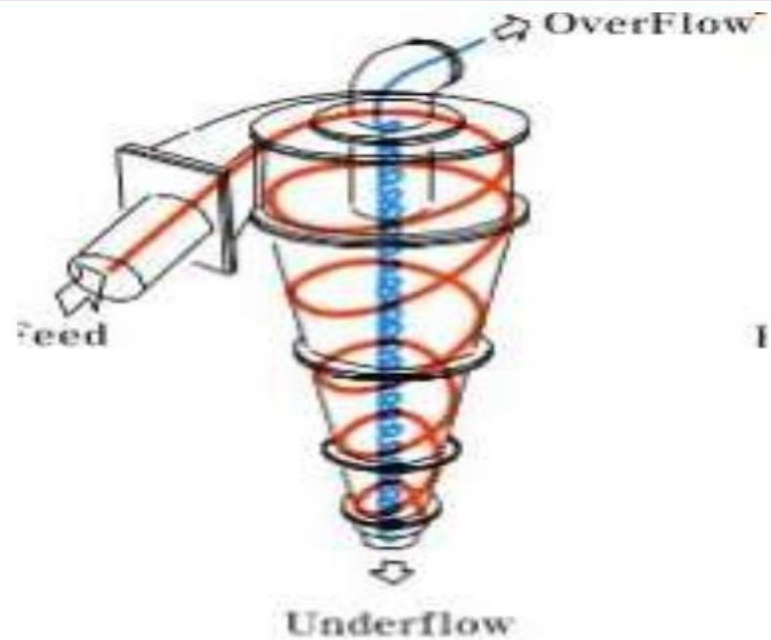


9.11
figure

Flow patterns in a gas-solid cyclone separator (adapted from Geankoplis, 1983).





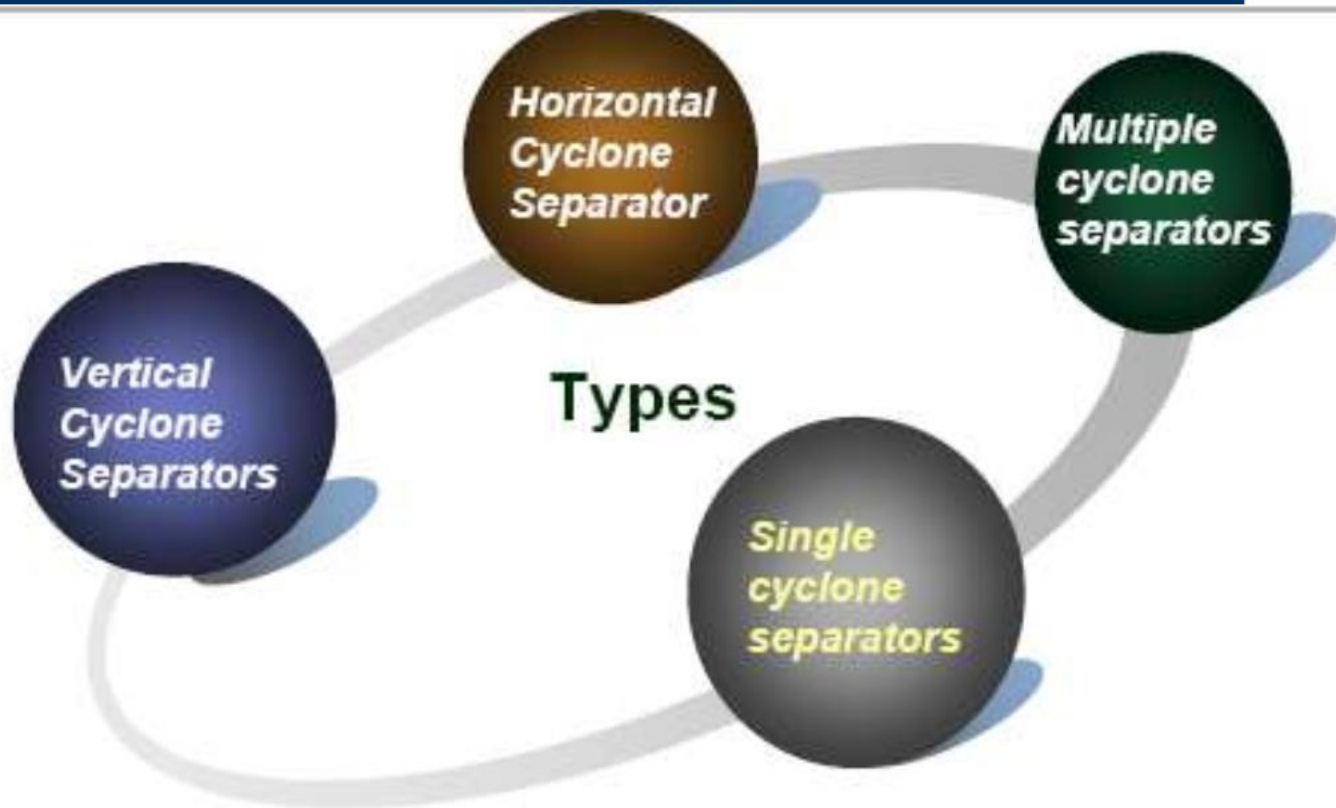


Simple Operation

Simple Maintenance

Simple Construction

Types



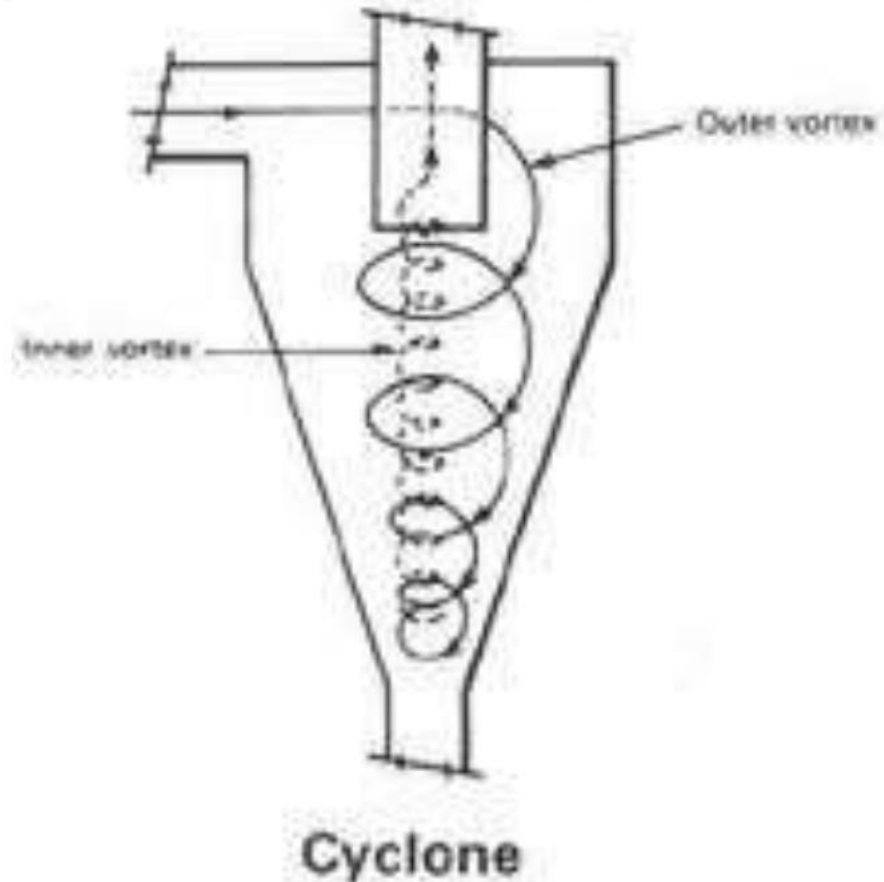
Vertical Cyclone Separator



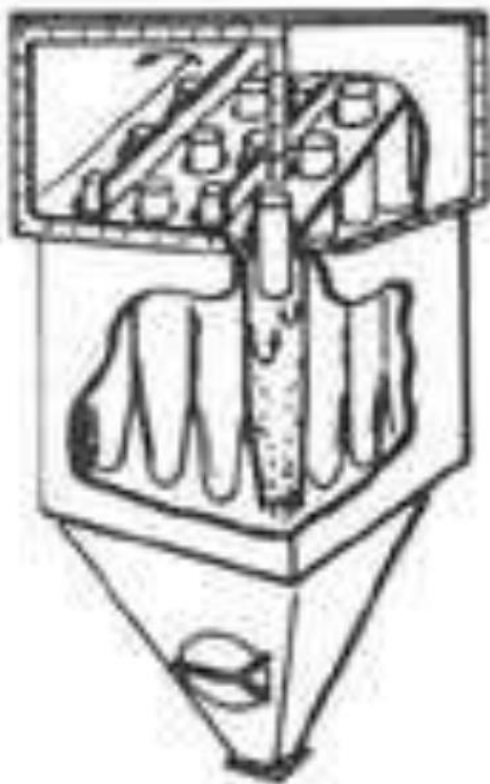
- Vertical cyclone separators are used to remove 99% of free liquids and solids 5 microns and larger.
- These separators are particularly effective for the removal of solids, such as iron sulphides. The high efficiency of these separators is due to the unique design which enables the separator to be approximately $\frac{1}{3}$ the diameter of a conventional separator.

Single-cyclone separators

- They create a dual vortex to separate coarse from fine dust.
- The main vortex spirals downward and carries most of the coarser dust particles.
- The inner vortex created near the bottom of the cyclone, spirals upward and carries finer dust particles.



Multi Cyclone separator



Multiclone

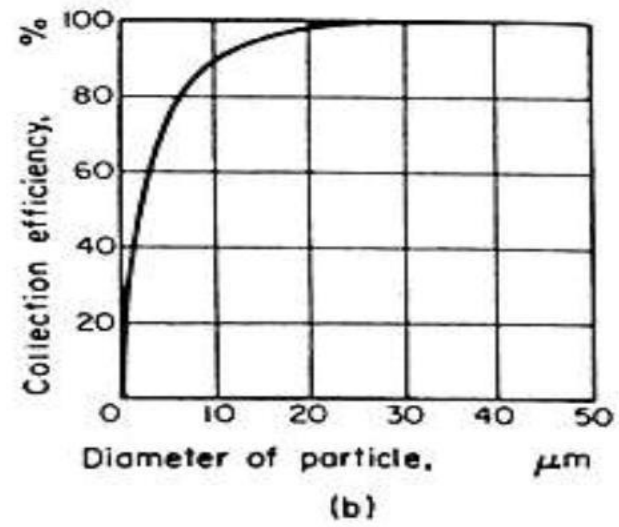
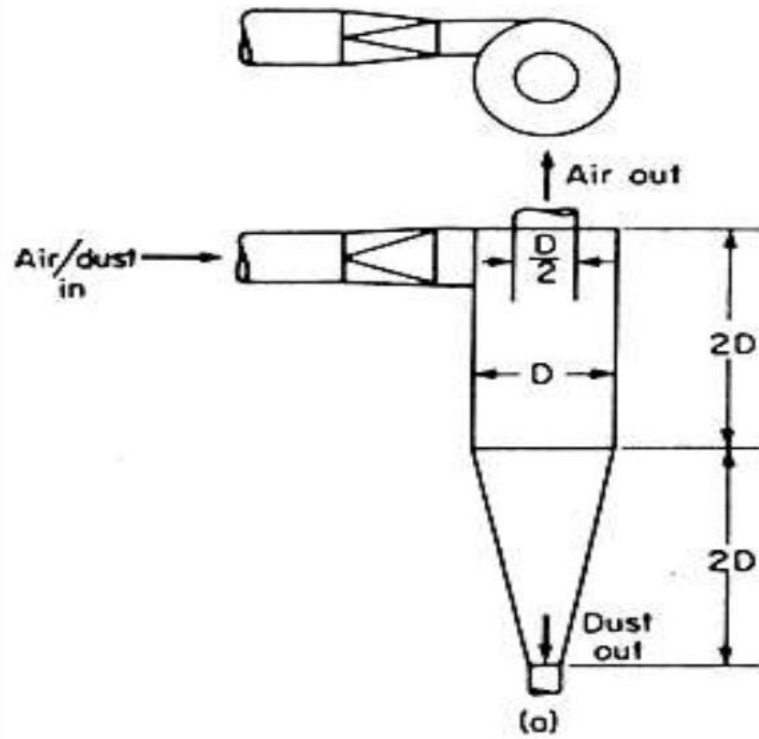
- It consists of a number of small-diameter cyclones, operating in parallel and having a common gas inlet and outlet, as shown in the figure.
- Multi-clones operate on the same principle as cyclones--creating a main downward vortex and an ascending inner vortex.
- Multi clones are more efficient than single cyclones because they are longer and smaller in diameter.
- The longer length provides longer residence time while the smaller diameter creates greater centrifugal force. These two factors result in better separation of dust particulates.

Design Procedure

- **The design parameters for the cyclone separator are as follows:**

- Length of cylinder $L_1 = 2D_2$
- Length of cone $L_2 = 2D_2$
- • Diameter of exit $D_e = 0.5D_2$
- • Height of entrance $H = 0.5D_2$
- • Width of entrance $B = 0.25D_2$
- • Diameter of dust exit $D_d = 0.25D_2$
- • Length of exit duct $L_3 = 0.125D_2$

SORUCE: Crawford 1976



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- The efficiency of collection of various particles (η) can be determined from the empirical relation and graph developed by Lapple 1951.

$$d_{0.5} = \left[\frac{9\mu B^2 H}{\rho_p Q_g \theta} \right]^{1/2}$$

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$d_{0.5}$	= cut diameter
μ	= Dynamic viscosity 'Pa.s'
B	= width of entrance 'm'
H	= Height of entrance 'm'
ρ	= Particle density 'kg/m ³ .s'
Q_g	= Gas flow rate m ³ /s
θ	= No of turns in cyclone

$$\theta = \frac{\pi}{H} (2 L_1 + L_2)$$

Cyclone Collector Design Considerations:

- **Particle size (particles with larger mass being subjected to greater force),**
- **Force exerted on the dust particles**
- **Time that the force is exerted on the particles**